

INTRODUCCION A LA CRIPTOGRAFIA MODERNA

Homework 3

Depto. de Computación, FCEN, UBA, 1er. Bimestre 2026

28/4/26

Student:

Due: 12/5/26, 23:59 hs

Instructions

- Upload your solution to Campus; make sure it's only one file, and clearly write your name on the first page. Name the file '<your last name>_HW3.pdf.'
If you are proficient with $\text{\LaTeX}$, you may also typeset your submission and submit in PDF format. To do so, uncomment the "`%\begin{solution}`" and "`%\end{solution}`" lines and write your solution between those two command lines.
- Your solutions will be graded on *correctness* and *clarity*. You should only submit work that you believe to be correct.
- You may collaborate with others on this problem set. However, you must **write up your own solutions** and **list your collaborators and any external sources (including ChatGPT and similar generative AI chatbots)** for each problem. Be ready to explain your solutions orally to a member of the course's staff if asked.

This assignment contains 6 questions, for a total of 60 points.

1. (10 points) Consider the following key-exchange protocol:

- (a) Alice chooses uniform $k, r \in \{0, 1\}^n$, and sends $s := k \oplus r$ to Bob.
- (b) Bob chooses uniform $t \in \{0, 1\}^n$, and sends $u := s \oplus t$ to Alice.
- (c) Alice computes $w := u \oplus r$ and sends w to Bob.
- (d) Alice outputs k and Bob outputs $w \oplus t$.

Show that Alice and Bob output the same key. Analyze the security of the scheme (i.e., either prove its security or show a concrete attack).

2. (10 points) Formally define the CDH assumption (i.e., first define a CDH experiment, and then use it to define what it means for the CDH problem to be hard). Prove that hardness of the CDH problem relative to $\mathcal{G}$ implies hardness of the discrete-logarithm problem relative to $\mathcal{G}$, and that hardness of the DDH problem relative to $\mathcal{G}$ implies hardness of the CDH problem relative to $\mathcal{G}$.
3. Suppose you know two El Gamal ciphertexts $\langle c_1, c_2 \rangle$ and $\langle c'_1, c'_2 \rangle$ that encrypt unknown plaintexts m and m' . You also know the public key h and cyclic group generator g .

- (a) (5 points) What information can you infer about m and m' if you observe that $c_1 = c'_1$?
 - (b) (5 points) What information can you infer about m and m' if you observe that $c_1 = g \cdot c'_1$?
4. (10 points) In class we showed that El Gamal encryption is *malleable*, and specifically that given a ciphertext $\langle c_1, c_2 \rangle$ that is the encryption of some unknown message m , it is possible to produce a ciphertext $\langle c_1, c'_2 \rangle$ that is the encryption of $\alpha \cdot m$ (for known α). A receiver who receives both these ciphertexts might be suspicious since both ciphertexts share the first component. Show that it is possible to generate $\langle c'_1, c'_2 \rangle$ that is the encryption of $\alpha \cdot m$, with $c'_1 \neq c_1$ and $c'_2 \neq c_2$.
 5. (10 points) Explain why the RSA exponents e and d must always be odd numbers.
 6. (10 points) Consider the RSA-based encryption scheme in which a user encrypts a message $m \in \{0, 1\}^\ell$ with respect to the public key $\langle N, e \rangle$ by computing $\tilde{m} := H(m) \parallel m$ and outputting the ciphertext $[\tilde{m}^e \bmod N]$. (Here, let $H : \{0, 1\}^\ell \rightarrow \{0, 1\}^n$ and assume $\ell + n < \|N\|$.) The receiver recovers $\tilde{m}$ in the usual way and verifies that it has the correct form before outputting the ℓ least-significant bits as m . Prove or disprove that this scheme is CCA-secure if H is modeled as a random oracle.